$$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$$

Vector $\begin{pmatrix} n_0(1) \\ n_1(1) \\ \vdots \\ n_{m-1}(1) \end{pmatrix}$ = Matrix $\begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix}$ Vector $\begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_{m-1}(0) \end{pmatrix}$

$\mathbf{n}(2)$ ← **L** ← $\mathbf{n}(1)$ ← **L** ← $\mathbf{n}(0)$

$$\begin{pmatrix} n_0(t+1) \\ n_1(t+1) \\ \vdots \\ n_{m-1}(t+1) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix} \begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \end{pmatrix}$$

$\mathbf{n}(t+1) = \mathbf{L}\mathbf{n}(t)$,

$\mathbf{L} \in m \times m$

$$\begin{aligned} \mathbf{L}\mathbf{u}_1 &= \lambda_1 \mathbf{u}_1 \\ \mathbf{L}\mathbf{u}_2 &= \lambda_2 \mathbf{u}_2 \\ &\vdots \\ \mathbf{L}\mathbf{u}_m &= \lambda_m \mathbf{u}_m \end{aligned}$$

Eigenvalue	Eigen Vector
λ_1	$\mathbf{u}_1$
λ_2	$\mathbf{u}_2$
$\vdots$	$\vdots$
λ_m	$\mathbf{u}_m$

$^*\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$

Diagonalization of matrix L

$$\mathbf{L} \Rightarrow \mathbf{U} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix}$$

$$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1} \qquad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$

$$\mathbf{L}^t = \underbrace{\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}}_{\mathbf{I}}\underbrace{\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}}_{\mathbf{I}}\underbrace{\mathbf{U}}_{\mathbf{I}}\cdots\underbrace{\mathbf{U}^{-1}\mathbf{U}}_{\mathbf{I}}\mathbf{\Lambda}\mathbf{U}^{-1} = \mathbf{U}\mathbf{\Lambda}^t\mathbf{U}^{-1}$$

$$\mathbf{\Lambda}^t = \begin{pmatrix} \lambda_1^t & & 0 \\ & \ddots & \\ 0 & & \lambda_m^t \end{pmatrix}$$

Apply what you have learned so far.



P115-118

$$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$$

$$= \mathbf{U}\mathbf{\Lambda}^t\mathbf{U}^{-1}\mathbf{n}(0) = \boxed{\mathbf{U}\mathbf{\Lambda}^t}\boxed{\mathbf{U}^{-1}\mathbf{n}(0)}$$

$$\mathbf{U}\mathbf{\Lambda}^t = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1^t & & 0 \\ & \ddots & \\ 0 & & \lambda_m^t \end{pmatrix} = (\lambda_1^t \mathbf{u}_1, \lambda_2^t \mathbf{u}_2, \cdots, \lambda_m^t \mathbf{u}_n)$$

$$\mathbf{U}^{-1}\mathbf{x}(0) = \begin{pmatrix} \blacksquare \end{pmatrix} \begin{pmatrix} \blacksquare \end{pmatrix}$$

$$(\lambda_1^t \mathbf{u}_1, \lambda_2^t \mathbf{u}_2, \cdots, \lambda_n^t \mathbf{u}_n) \begin{pmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{pmatrix} = g_1\lambda_1^t\mathbf{u}_1 + g_2\lambda_2^t\mathbf{u}_2 + \cdots + g_n\lambda_n^t\mathbf{u}_n$$



$$\underbrace{\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \end{pmatrix}}_{\mathbf{n}(t)} = \underbrace{\begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix}^t}_{\mathbf{L}^t} \underbrace{\begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_{m-1}(0) \end{pmatrix}}_{\mathbf{n}(0)}$$

Determine the eigenvalues and the eigenvectors of matrix **L**

Eigenvalue	Eigen Vector
λ_1	$\mathbf{u}_1$
λ_2	$\mathbf{u}_2$
$\vdots$	$\vdots$
λ_m	$\mathbf{u}_m$

$^*\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$

$$\mathbf{n}(t) = \mathbf{L}^t\mathbf{n}(0)$$

$$= g_1\lambda_1^t \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix} + g_2\lambda_2^t \begin{pmatrix} | \\ \mathbf{u}_2 \\ | \end{pmatrix} + \cdots + g_m\lambda_m^t \begin{pmatrix} | \\ \mathbf{u}_m \\ | \end{pmatrix}$$

Life table

age (x)	l_x	s_x	b_x
0	1.	0.5	
1	0.5	0.6	0.
2	0.3	0.1	1.67
3	0.03	0.	20.00
4	0.		

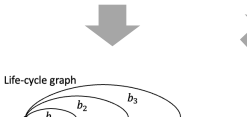
$l_0 = 1$

$l_1 = s_0 l_0$

$l_2 = s_1 l_1 = s_0 s_1 l_1$

...

$$\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \\ n_m(t) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}^t \begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_{m-1}(0) \\ n_m(0) \end{pmatrix}$$



$$\mathbf{n}(t) = \mathbf{L}^t\mathbf{n}(0)$$

Why?

$$\mathbf{n}(t) = g_1\lambda_1^t\mathbf{u}_1 + g_2\lambda_2^t\mathbf{u}_2 + \cdots + g_m\lambda_m^t\mathbf{u}_m$$

$$\mathbf{n}(t) = g_1\lambda_1^t \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix} + g_2\lambda_2^t \begin{pmatrix} | \\ \mathbf{u}_2 \\ | \end{pmatrix} + \cdots + g_m\lambda_m^t \begin{pmatrix} | \\ \mathbf{u}_m \\ | \end{pmatrix}$$

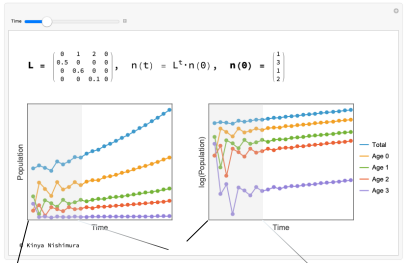
$$\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \\ n_m(t) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & 0 & \cdots & 0 \\ 0 & s_1 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}^t \begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_{m-1}(0) \\ n_m(0) \end{pmatrix}$$

$$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$$

$$= g_1 \lambda_1^t \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix} + g_2 \lambda_2^t \begin{pmatrix} | \\ \mathbf{u}_2 \\ | \end{pmatrix} + \cdots + g_m \lambda_m^t \begin{pmatrix} | \\ \mathbf{u}_m \\ | \end{pmatrix}$$

$$* \ \lambda_1 \geq \lambda_2 \geq \cdots$$

$$= \lambda_1^t \left(g_1 \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix} + g_2 \underbrace{\left(\frac{\lambda_2}{\lambda_1}\right)^t}_{\to 0} \begin{pmatrix} | \\ \mathbf{u}_2 \\ | \end{pmatrix} + \cdots + g_m \underbrace{\left(\frac{\lambda_m}{\lambda_1}\right)^t}_{\to 0} \begin{pmatrix} | \\ \mathbf{u}_m \\ | \end{pmatrix} \right) \approx g_1 \lambda_1^t \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix}$$



$$g_1 \lambda_1^t \begin{pmatrix} | \\ \mathbf{u}_1 \\ | \end{pmatrix} + g_1 \lambda_2^t \begin{pmatrix} | \\ \mathbf{u}_2 \\ | \end{pmatrix} + \cdots + g_1 \lambda_m^t \begin{pmatrix} | \\ \mathbf{u}_m \\ | \end{pmatrix}$$

Leslie_Projection_Model_highcontras

Leslie matrix : $\mathbf{L}(s_i, b_i), \ (i = 0, 1, \ldots)$

Dominant eigenvalue

Dominant eigenvector

$$\mathbf{n}_t \propto \lambda^t \mathbf{u} \propto \begin{pmatrix} n_0 \\ n_1 \\ \vdots \end{pmatrix}_t =$$

Senior

Young

Rate of increase

Kept regardless of time, t : Stable-age distribution

+ Cultural factors

educational economical religious public hearth

+ Historical Events

+ External and Internal Change

Real Complex

Math models

Simple

$$m\frac{d^2x}{dt^2} = mg - C_D \rho S \left(\frac{dx}{dt}\right)^2 + \cdots$$

$$\mathbf{n}_t \propto \lambda^t \mathbf{u}$$

$$m\frac{d^2x}{dt^2} = mg$$

Further Thinking Simple

The **fundamental** properties of a population resulting from the nature of individuals, reproduction and survival (death).

✓ A rule of population **increase**

✓ A rule of population **structure**

$$\mathbf{n}_t \propto \lambda^t \mathbf{u}$$

Real Complex

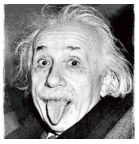
Math models

Simple

$$m\frac{d^2x}{dt^2} = mg - C_D \rho S \left(\frac{dx}{dt}\right)^2 + \cdots$$

$$m\frac{d^2x}{dt^2} = mg$$

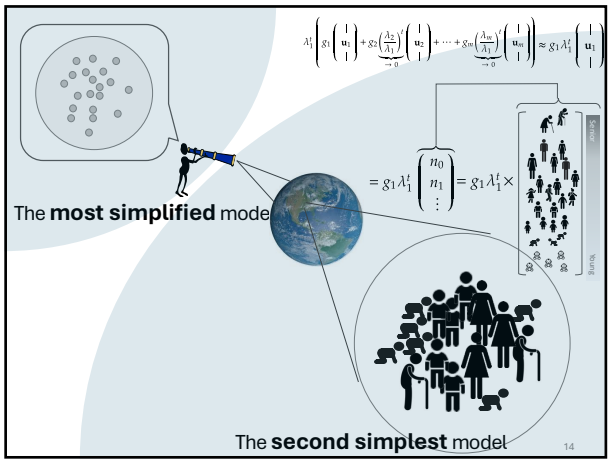
12



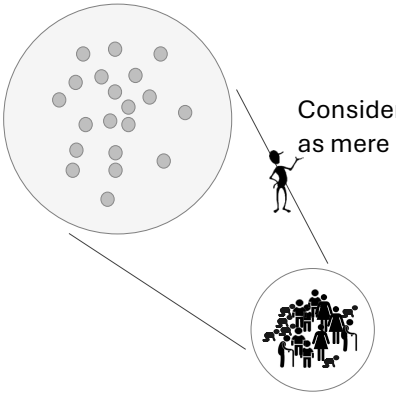
To find a principle, think and investigate in a simplified way.

Everything must be made as simple as possible.

However, we must not oversimplify.



The **most simplified** view



Consider the individuals as mere particles.

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$$N_{t+1} = N_t + \underbrace{B_t}_{bN_t} - \underbrace{D_t}_{dN_t}$$

b: per capita birth rate/unit time

d: death rate/unit time

$$N_{t+1} = N_t + bN_t - dN_t$$

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$$N_{t+1} = N_t + bN_t - dN_t$$

$$= \underbrace{(1 - d)}_{s} N_t + bN_t$$

s: survival rate

$$N_{t+1} = \underbrace{sN_t}_{\text{survivors}} + \underbrace{bN_t}_{\text{newborn}}$$

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$$N_{t+1} = \overbrace{sN_t}^{\text{survivor}} + \overbrace{bN_t}^{\text{Newborn}}$$

$$N_{t+1} = \underbrace{(s + b)}_{\lambda} N_t$$

$$\frac{N_{t+1}}{N_t} = \lambda$$: growth rate per one period

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Population | Individual

Change	Mechanism	
$\frac{N_{t+1}}{N_t} = \underbrace{\lambda}_{s+b}$	$= (1-d)+b$ $= b-d+1$ $= R+1$	*b*, *d* or *s*

We see the same thing differently.

$N_t = \lambda^t N_0$

$$\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_{m-1}(t) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-2} b_{m-1} & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & 0 & s_{m-2} & 0 \end{pmatrix}^t \begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_{m-1}(0) \end{pmatrix}$$

$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$

The formulas look different.

How are they different?
What does the difference mean?

What do they have in common?
What do commonalities imply?

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$N_t = \lambda^t N_0$

$g_1\lambda_1^t \begin{pmatrix} \\ \mathbf{u}_1 \\ \\ \end{pmatrix} + g_2\lambda_2^t \begin{pmatrix} \\ \mathbf{u}_2 \\ \\ \end{pmatrix} + \cdots + g_m\lambda_m^t \begin{pmatrix} \\ \mathbf{u}_m \\ \\ \end{pmatrix} \quad \cdots$

	No information about age	Incorporate attribute of age
Change	Geometric	Geometric + zigzag
Structure	—	Invariant + transient

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+ Cultural factors
educational
economical
religious
public hearth

+ Historical Events

+ External and Internal Change

$\mathbf{n}_t \propto \lambda^t \mathbf{u}$

Real — Complex
Math models
Simple

Key Messages to Emphasize

Don't say the real world isn't like that.
Notice that we now think about an **ideal population**.

0. Introduction
1. Nature of individual life: Death process
2. Nature of individual life: Childbearing
3. Characteristics of population that arise from the nature of individual's life
4. Traits of individuals defined in the population
5. Synthesis of the interdisciplinary science

Structure

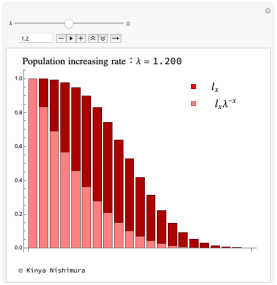
Size $\propto \lambda^t$,

Stable age composition

$$\propto \mathbf{u} = \begin{pmatrix} l_0\lambda^{-0} \\ l_1\lambda^{-1} \\ l_2\lambda^{-2} \\ l_3\lambda^{-3} \\ \vdots \\ l_m\lambda^{-m} \end{pmatrix}$$

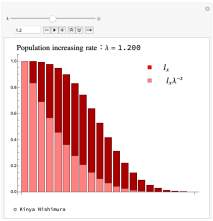
I call the population as "Ideal Population"

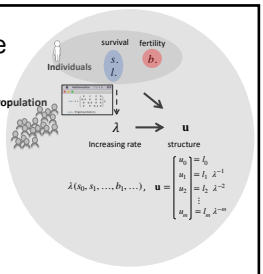
Hidden beneath real population, yet always at work — the most fundamental property of a population.



StableAgeDistribution_highcontrast.nb

In summary: thinking time

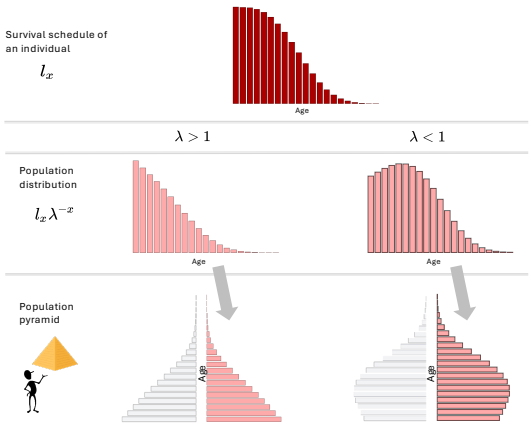




✓ How is the relationship between survival curve, and the stable age distribution curve?

StableAgeDistribution_highcontrast.nb

✓ How do you interpret the relationship between survival curve, and the stable age distribution curve?



Ideal Population

at any arbitrary year

Relative population size by age	Age-specific fecundity
$u_0 = \lambda^{-0} l_0$	b_0
$u_1 = \lambda^{-1} l_1$	b_1
$u_2 = \lambda^{-2} l_2$	b_2
$\vdots$	$\vdots$
$u_m = \lambda^{-m} l_m$	b_m

Quiz

What is the average age of the population?

What age would be representative of this population?

✓ What is the formula to get that age by using the equations?

What is the average age at childbearing for this group?

What is the average age of the mothers who represent this group?

✓ What is the formula to get that age by using the equations?

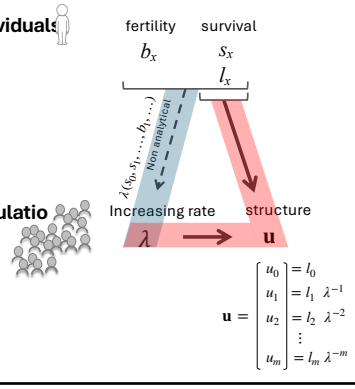
Relative population size by age	Age-specific fecundity
$u_0 = \lambda^{-0} l_0$	b_0
$u_1 = \lambda^{-1} l_1$	b_1
$u_2 = \lambda^{-2} l_2$	b_2
$\vdots$	$\vdots$
$u_m = \lambda^{-m} l_m$	b_m

$$p(x) = \frac{b_x \lambda^{-x} l_x}{\sum_{s=0}^{\infty} b_s \lambda^{-s} l_s}$$

The average age of childbirth of the population

$$\sum_{x=0}^{\infty} x p(x) = \sum_{x=0}^{\infty} x \underbrace{\frac{b_x \lambda^{-x} l_x}{\sum_{s=0}^{\infty} b_s \lambda^{-s} l_s}}_{p(x)} = \frac{\sum_{x=0}^{\infty} x\, b_x \lambda^{-x} l_x}{\sum_{s=0}^{\infty} b_s \lambda^{-s} l_s}$$

The fundamental relationship between the individuals' demographic parameters and the population parameters



$$\mathbf{u} = \begin{bmatrix} u_0 = l_0 \\ u_1 = l_1 \lambda^{-1} \\ u_2 = l_2 \lambda^{-2} \\ \vdots \\ u_m = l_m \lambda^{-m} \end{bmatrix}$$

